



Large-language-model-driven agents for fire evacuation simulation in a cellular automata environment

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ARTICLE INFO

Keywords:

Large language model
Fire evacuation
Multi agent
Cellular automata

ABSTRACT

This study presents a multi-agent fire evacuation model driven by large language models (LLMs), aiming to simulate human-like decision-making and behaviors during fire emergencies. The proposed model endows agents with personalized memory, cognition, and decision-making capabilities through LLMs, while characterizing the evacuation environment using a combination of spatial semantics and cellular automata. This approach enables agents to perceive, decide, and move within a unified spatiotemporal framework. The model is tested using various LLMs with different scales, including ChatGPT 4.0, ERNIE-Bot 4.0, Llama-2-70B-Chat, and ChatGLM2-6B-32K, in a simulated shopping mall fire evacuation scenario constructed from LiDAR scans and 3D reconstruction. Experimental results demonstrate that the LLM-driven agents exhibit exploratory and adaptive behaviors consistent with real-world scenarios, with larger-scale LLMs generating more consistent and efficient evacuation strategies. The study also reveals the impact of agent background settings and communication behaviors on evacuation outcomes. This research contributes to the advancement of intelligent agent-based modeling for emergency evacuation simulations and provides insights into the potential of LLMs in enhancing the realism and effectiveness of such models. Future work should focus on increasing scenario complexity, model interpretability, and controllability to better serve real-world evacuation management and decision-making.

1. Introduction

With the increase in population and the growing complexity of buildings, fire evacuation simulation has become particularly important. Utilizing mathematical and computational tools, these models are capable of simulating the evacuation behavior of individuals during a fire, detailing the movement of people, building structures, and the spread of fire. Through quantitative analysis, fire evacuation simulation can assist decision-makers in optimizing evacuation routes, allocating emergency resources reasonably, and providing a basis for the formulation of emergency plans. These models play a crucial role in disaster prevention, mitigation, and enhancing public safety.

Currently, fire evacuation models can be categorized into three types: macroscopic models, microscopic models, and hybrid models (Ronchi et al., 2019; Chen et al., 2021; Kaur and Kaur, 2022). Macroscopic models simulate crowd movement using hydrodynamic equations or continuum theory, focusing on overall behavior. For example, Hughes (2002) modeled pedestrian flow akin to the Navier-Stokes equations,

while Colombo and Rosini (2005) analyzed high-density crowd dynamics with shock wave theory. However, these models struggle to capture individual behaviors and micro-interactions due to their neglect of individual heterogeneity, such as decision-making and local interactions (Li et al., 2018). Microscopic models simulate individual attributes—such as movement speed, target location, and decision-making logic—along with their interactions, including environmental responses and behaviors like avoidance and following, to achieve bottom-up emergence of group behavior. Typical approaches include the Social Force Model and Agent-Based Models (Helbing & Molnár, 1995; Pan et al., 2007). However, these models require significant computational resources in large-scale scenarios (Dang et al., 2021). The hybrid model integrates the strengths of both macro and micro levels, striving to strike a balance between the two, and is suitable for complex evacuation scenarios involving multi-scale interactions (Tissera et al., 2012; Hassannayebi et al., 2022). For instance, Xiong et al. (2022) combined the continuum model with the cellular automaton model, which not only reflects the overall movement trends of crowds but also

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<https://doi.org/10.1016/j.ssci.2025.106935>

Received 12 October 2024; Received in revised form 1 June 2025; Accepted 21 June 2025

Available online 7 July 2025

0925-7535/© 2025 Published by Elsevier Ltd.

captures the diverse behaviors of individuals. However, such models must also overcome the inherent limitations of each approach.

With the development of artificial intelligence technology, computational power and efficiency have significantly improved, making it easier for researchers to conduct studies on microscopic algorithms. An Agent is a type of microscopic model that simulates human attributes and behavior, making decisions based on the environment (Shi et al., 2009). Agent-based models simulate human behavior in real-world environments by defining attributes, behaviors, and mechanisms (Gilbert and Terna, 2000; Pan et al., 2007). Researchers hope to achieve swarm intelligence by endowing Agents with more elements, making the Agent-based model exhibit behavior similar to humans (Li et al., 2018). Despite the success of Agent-based models, there are still limitations, such as the complexity of the system due to considering too many elements, and the difficulty in quantifying and simulating human emotions, psychology, and cultural factors (LEE & Malkawi, 2013).

In recent years, LLMs based on the Transformer architecture have demonstrated groundbreaking progress in cognitive tasks. Models like ChatGPT and Claude3.5, trained on massive text corpora, have not only acquired factual memory capabilities but also exhibited decision-making levels comparable to humans in reasoning tasks (Aidan Gilson et al., 2023; Zhu et al., 2024a). Current research has begun exploring the potential of LLMs in fire evacuation decision support, such as optimizing dynamic path planning (Dang et al., 2024; Durmus et al., 2024; Luo et al., 2025) and analyzing social media behavioral data (Wu et al., 2025). However, existing studies primarily focus on technical adaptations at the task execution level and have yet to systematically construct an LLM-based human cognition-behavior fire evacuation simulation framework, lacking dynamic simulations of individual risk perception and behavioral decision-making. Meanwhile, LLMs have already demonstrated human-like thought patterns in other tasks (Yoo and Lee, 2023; Webb et al., 2023), providing a theoretical foundation for building fire evacuation agent models. Nevertheless, translating this potential into practical applications still faces challenges: how to enable LLMs to perceive fire evacuation scenarios through language descriptions and how to simulate interactions between individuals and between individuals and the environment.

In response to these issues, we have designed an Agent-based fire evacuation model built upon LLMs, aiming to create credible intelligent agents that simulate individuals with varying attributes during a fire evacuation. These agents possess independent physiological attributes, memory, and decision-making capabilities, and can communicate with other agents to collaboratively complete fire evacuation tasks.

2. Background

2.1. Agent-based fire evacuation model

Agent-based fire evacuation models are computational methods that utilize agent technology to simulate and analyze the evacuation process of people in fire scenarios. In these algorithms, each evacuee is considered an intelligent agent with autonomous decision-making capabilities, making decisions based on their perception of the environment, interactions with other agents, and self-awareness. Multi-Agent Systems (MAS) allow for a bottom-up approach to evacuation, where the collective evacuation outcome is derived from simulating individual behaviors, a capability beyond the reach of macroscopic evacuation algorithms. Agent models are computation-intensive algorithms that were challenging to implement on traditional computing systems, but recent advancements in GPU computing chips and parallel computing technologies have made it possible to develop more complex Agent models.

Agent models typically require three elements: first, an Agent is an autonomous entity with its own needs, reactivity, proactivity, and social behaviors (Jennings & Wooldridge, 1998). Second, they must interact with the environment, sensing and altering it. Third, they should

communicate and interact with other Agents. These elements make Agents' behavior and decision-making in fire evacuation more human-like, aiming to simulate humans. Research on Agents spans diverse areas such as crowd behavior simulation (Helbing et al., 2000; Lovreglio et al., 2016), evacuation environment modeling (Pelechano & Badler, 2006; Chen et al., 2006), social interactions (Moussaïd et al., 2010), real-time simulation (Zheng et al., 2009), and virtual reality (Ronchi et al., 2016; Dang et al., 2021). These studies significantly contribute to Agent construction, environmental complexity, method integration, and scenario diversity. However, limitations exist as Agents' behaviors are predefined, and simulating all human behaviors is unattainable. Despite progress with deep learning (Chen & Xue, 2015; Zhao et al., 2020; Hou et al., 2022; Zhu et al., 2023), a comprehensive method to simulate human behavior is lacking. Systems that fully simulate human behavior or thought are termed Artificial General Intelligence (AGI), defined as AI systems with human-like intelligence (Minsky, 1961; Turing, 2009; Legg and Hutter, 2007). Recent emergent capabilities of LLMs show significant progress towards AGI, exhibiting capabilities that approach or surpass human abilities in some tasks (Wu et al., 2023), offering a promising method for Agent-based fire evacuation simulation.

2.2. Multi-agent evacuation systems

MAS consist of multiple intelligent agents capable of acting independently and cooperating to achieve objectives (Tampuu et al., 2015). In MAS, each agent possesses a certain degree of autonomy, enabling self-initiated decisions and actions. Key features of MAS include autonomy, local perspectives, decentralization, and the capacity for both collaboration and competition. Within fire evacuation simulations, the characteristics of MAS can be further defined as follows: each agent represents an independent evacuee controlling their own state and behavior; each agent only accesses local information around them or obtains information through communication with other agents; decision-making is decentralized among agents without a single control point; agents may collaborate and compete during evacuation (Sharma et al., 2018).

The application and study of MAS in evacuation scenarios are becoming increasingly significant, especially in simulating and managing crowd evacuations during emergencies. Research includes integrating MAS with the Internet of Things to recommend the most efficient evacuation routes (Neto et al., 2019); expanding MAS parameters to optimize evacuation processes (Zhang & Shen, 2019); incorporating robots into evacuation processes and assessing their impact on evacuation with MAS (Uchiya et al., 2019); and combining MAS with fuzzy logic to enhance the realism and effectiveness of simulated individual evacuations (Sahin et al., 2019). These studies aim to improve the realism of evacuations using MAS or to extend MAS to accommodate more evacuation scenarios. Despite significant progress, these approaches are mostly tailored for specific evacuation scenarios and require additional parameters to control evacuation processes for more realistic individual behaviors, making it challenging for MAS to balance between parameter (or rule) specificity and universality. Adopting reasoning methods can effectively reduce the number of parameters and rules, and LLMs offer potential solutions to this issue.

3. Method

3.1. Agent memory and cognitive constraints

Memory serves as the foundation for humanoid agents to simulate human behavior and decision-making, supporting the accumulation of experience and behavioral evolution (Zhu et al., 2024b), and providing data support for learning adaptability (Dang et al., 2023). In multi-agent fire evacuation scenarios, agent interactions rely on memory construction. However, while LLMs can utilize pre-trained knowledge to process general information, they struggle to simulate individual specificity (e.

g., age differences). To address this, two strategies are needed: 1) prompt constraints (e.g., “simulate the decision-making of an 8-year-old child”) to limit the model’s output range; and 2) task fine-tuning, optimizing the model through training with specific data. Due to resource efficiency, prompt constraints are more practical.

The conversational mechanism of LLMs limits the context of user input. Despite expanding context length, research indicates that overly long contexts can cause LLMs to overlook details, leading to Catastrophic Forgetting (Zhai et al., 2023). To model Agents with different characteristics, we divide an Agent’s memory into short-term and long-term memory. Short-term memory, akin to human working memory, consists of real-time environmental information (building layout, fire location, etc.), personal status (current location and direction of movement of the Agent, etc.), and social dynamics information (behavior of surrounding crowd, contact with other Agents, etc.), submitted to the LLM in the form of prompts. Considering these inputs continuously feed into the LLM during fire evacuation, past information remains relevant to current decisions, necessitating its storage in a time-series format and submission as prompts to the LLM. Short-term memory primarily processes evacuation environment information, while long-term memory shapes the Agent’s personality traits, such as individual physiological attributes, familiar routes in evacuation settings, specific events, and preferences. Not all this information is submitted as prompts in every dialogue round with the LLM but is queried actively or loaded into prompts under specific conditions.

To implement this mechanism, we employ OpenAI’s text-embedding-3-small model for text embedding to manage agent long-term memory storage and retrieval. The workflow proceeds as follows: Long-term memories are first segmented into fixed-length 512-token text fragments through a sliding window approach, with consecutive fragments maintaining a 128-token overlap to preserve contextual continuity. Each fragment is then encoded into a 1536-dimensional vector representation using text-embedding-3-small, subsequently stored in the database. Leveraging the model’s proven semantic representation capabilities in high-dimensional spaces, query statements undergo identical encoding followed by nearest-neighbor search via cosine similarity measurement, with a confidence threshold of 0.75. To address the inherent semantic sparsity yet content relevance among segmented sentences, we organize long-term memory into multi-vector memory chunks. When any constituent vector within a chunk demonstrates sufficient semantic affinity with input sentences, the entire memory chunk is retrieved as contextual prompts for LLMs, as illustrated in Fig. 1.

The pseudocode for this process is shown in Algorithm 1.

Algorithm 1. (Embedding and retrieval of long-term memory)

```
Initialize Simulation (Grid, Agents, DecisionInterval  $\Delta_t$ )
time = 0
WHILE simulation_running DO
  IF time MOD  $\Delta_t$  == 0 THEN
    FOR EACH Agent DO
```

(continued on next column)

(continued)

```
  Perceive Environment
  Decision = Query LLM (Perception + History)
  Update History
  Update Agent Intention Based On Decision
END FOR
END IF
FOR EACH Agent DO
  Act Based On Intention
END FOR
Update Environment State
Increment time
END WHILE
```

Using memory to simulate human cognition and behavior is an implicit method. For instance, an LLM can infer that “I am an 8-year-old child” may not independently choose a reasonable exit through implicit reasoning. However, to ensure Agents perform specific actions, we have incorporated explicit cognitive constraints into them. These constraints are formed by texts such as “You remember there is an emergency exit next to the elevator” and are present in each round of prompts.

It’s noteworthy that as evacuation time increases, a substantial amount of process information is injected into the Prompt, which not only exceeds the context limitations of LLMs but also makes it challenging for LLMs to reason based on a vast amount of text. In fact, humans adopt strategies of filtering and summarizing when dealing with a large volume of information, allowing more focus on current events. In fire evacuations, evacuation instructions, fire information, and exit information are the most critical, while the reasons behind each decision during the evacuation process are secondary. Therefore, we use LLMs to compress the historical memory of Agents, ensuring that the prompts remain concise while retaining key evacuation information, as shown in Fig. 2.

To optimize context management in prompt engineering, this study proposes a hierarchical information organization approach that structures original prompts into three distinct layers: contextual information, action information, and instruction information. The action information layer incorporates critical data elements including fire dynamics, evacuation decision-making, and environmental perception. To mitigate the performance degradation of LLMs caused by extended context lengths, a selective compression strategy is implemented, prioritizing retention of core information such as positional trajectories, environmental state changes, decision rationale, and chronological fire progression patterns. This structured information architecture not only effectively reduces the risk of memory loss in LLMs during long-context processing but also prevents agent behavioral loops through preserved trajectory data. Continuous monitoring of fire development dynamics further enhances decision-making capabilities by providing agents with comprehensive situational awareness, thereby improving the rationality and effectiveness of evacuation strategies.

Furthermore, this study employs a systematic prompt engineering approach in the interaction process based on LLMs. To ensure the

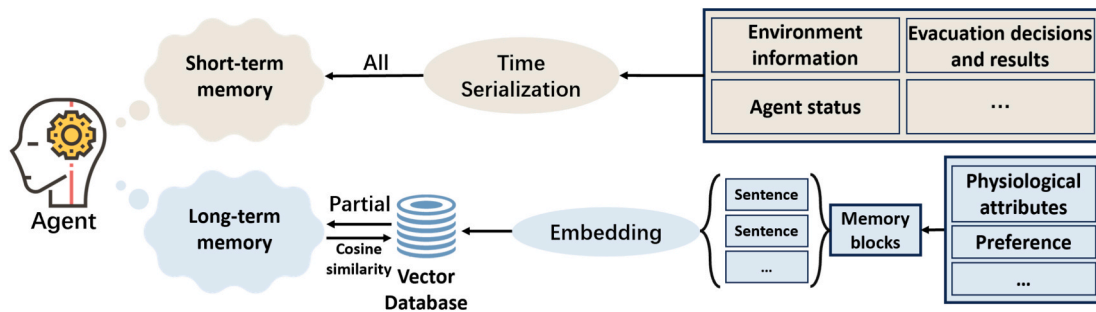


Fig. 1. Design of agent’s long-term and short-term memory.

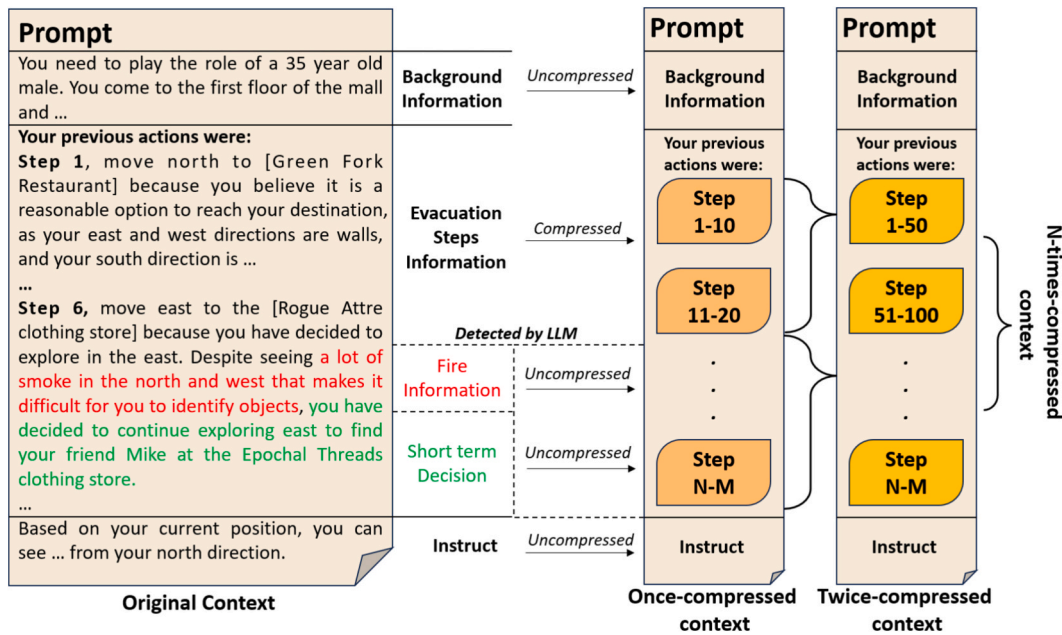


Fig. 2. The compression process of prompts.

effectiveness and parsability of the model's outputs, we have implemented the following core strategies in the design of prompts: 1) Clearly defining the task objectives and expected output results, guiding the model to focus on key issues through structured instructions; 2) Strictly standardizing the output format to JSON data structure, establishing a standardized response template to enhance the accuracy of computer program parsing; 3) Introducing the Few-shot learning mechanism, embedding complete examples in each task prompt to guide the model to follow the preset reasoning path and output specifications through demonstrative samples (Schulhoff et al., 2024).

3.2. Characterization of evacuation environments and autonomous actions

3.2.1. Cell-semantic hybrid modeling for evacuation environments

Agents' autonomous actions depend on their memory and perception of the environment. The method in 3.1 provides Agents with human-like memory; however, LLMs cannot directly perceive fire evacuation environments, necessitating a representation of the evacuation environment to the LLM. Cellular automata, as a mathematical model discrete in both time and space, offer precise spatial positioning and layout representation, generating a more accurate descriptive text of the environment. Furthermore, Agents' actions during evacuation require a spatiotemporal framework consistent with the environment, making cellular automata an effective method for environment characterization and constructing an action framework for Agents.

In our study, we describe evacuation environments using a spatial-semantic-cellular approach. Fire evacuation scenarios include multiple environmental objects (such as walls, tables, chairs, etc.), each occupying certain spaces. Thus, an environmental object should encompass both semantic information and spatial information represented by cellular automata. When characterizing environmental objects using cellular automata and semantics, two considerations are necessary: the multiplicity of semantics, as cellular automata are a two-dimensional structure, but evacuation environments are three-dimensional, inevitably leading to a cell having multiple spatial semantics, as shown in Fig. 3.

Visibility relationships in evacuation scenarios require consideration of the occlusion relationships, field of view, and inference thresholds between objects (cells). Human visual cognition of an object can be

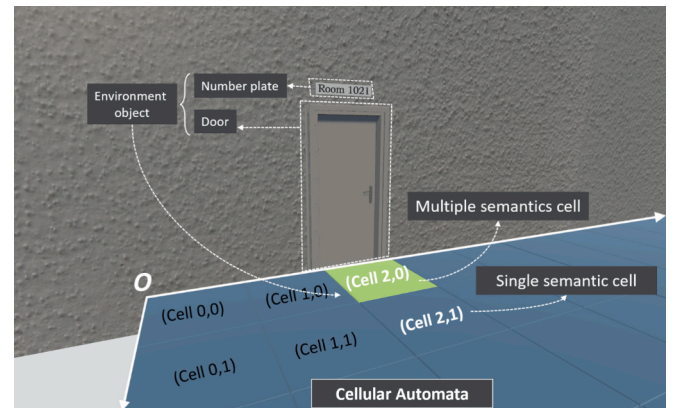


Fig. 3. Cell semantics and multiplicity of semantics.

summarized in three stages: vague perception, preliminary cognition, and full cognition (Man and Vision, 1982; Song and Zhao, 2025). In the vague perception stage, humans can only vaguely perceive the existence and rough shape of an object; in the preliminary cognition stage, they can recognize basic features of the object, such as color and shape. In the full cognition stage, humans can clearly understand the object-related information, including texture, expressed information, etc. Similarly, in evacuation scenarios, the distance between an object and an Agent is a key factor affecting the Agent's acquisition of object information. In our research, this process is simplified into Perceptual Threshold Distance and Full Perception Distance. At the Perceptual Threshold Distance, we input basic shape information about the object to the Agent and use terms like "seems" and "looks like" to indicate what it might be. At the Full Perception Distance, we provide the Agent with all information about the object, including its shape, color, semantics, contained text, etc. Cellular automata employ a two-dimensional grid representation; however, in actual evacuation environments, the height of objects also affects Agents' perception of the environment. Therefore, we assign height information to each cell and calculate the occlusion relationships between different cells.

Considering that Agents need to make movement decisions after perceiving the environment, we divide the Agent's field of view into

multiple directions and provide the LLM with prompts corresponding to those areas. In the decision-making phase, the Agent will decide the direction and position to move based on the descriptions of objects seen in different directions, as shown in Fig. 4.

To calculate the visible cells in a certain direction for an Agent, it's necessary to compute the cells within a certain angle and their occlusion relationships. We employed three steps: linear interpolation, stepwise detection, and occlusion detection. Assume the Agent's cell coordinates are $A(x,y)$, and the line-of-sight cell coordinates are $B(m,n)$. The increments in the x and y directions are calculated based on Formula 1.

$$\begin{aligned} x_{\text{increment}} &= \frac{dx}{\text{steps}} \# \\ y_{\text{increment}} &= \frac{dy}{\text{steps}} \# \end{aligned} \quad (1)$$

where $dx = m - x$ and $dy = n - y$ represent the difference in distance between the endpoint and the starting point along the x-axis and y-axis, respectively. The number of steps is calculated based on the maximum distance difference to ensure continuity of points along the path. The coordinates of each intermediate point $P(i)$ on the path are then calculated step by step according to the increment formula, as shown in Formula 2.

$$\begin{aligned} P_x(i) &= x + i * x_{\text{increment}} \# \\ P_y(i) &= y + i * y_{\text{increment}} \# \end{aligned} \quad (2)$$

Here, i iterates from 0 to steps, representing each step along the straight-line path. Finally, each point $P(i)$ on the path is checked to see if it exists in the set of impassable cells, blockedCells, as shown in Formula 3.

$$\text{blocked} = (P_x(i), P_y(i)) \in \text{blockedCells} \# \quad (3)$$

If any point $P(i)$ is blocked, the line of sight is considered obstructed; otherwise, the line of sight is unobstructed, as illustrated in Fig. 5.

Furthermore, we added a unified spatial reference for Agents, generating direction predicates about other cells based on their current position. For example: "You have previously visited [Room 503's door] located to your east, and currently, you can see [a red door] in your north direction." This helps the Agent understand past orientations and the direction of the next move.

3.2.2. LLM-agent action framework

The agent's perception of the environment depends on the cellular automaton's representation of that environment, while the agent's actions are also executed within the cellular automaton framework. This unified framework ensures that the agent's perception and actions share the same spatiotemporal context. Traditionally, an agent's next

movement position considers Moore or von Neumann neighborhoods. We utilize the cellular automaton's environmental representation to generate descriptive texts for 8 (or 4) directions, which are then submitted to the LLM for next-step action decisions. These textual descriptions are updated after each step. To enable the agent to comprehend the entire evacuation process, we record each step of its decision-making as new prompts submitted to the LLM for reference.

Before each agent movement operation, the system integrates two critical components into the prompt: (1) the structured textual representation of environmental information as described in Section 3.2.1, and (2) the compressed historical movement data from Section 3.1. This contextual integration allows the agent to understand its historical decisions and current status, thereby ensuring consistency in decision-making behavior.

If each movement of multi-agents (moving to adjacent cells) requires a perception-decision process, this would generate an excessive number of LLM dialogues during evacuation simulations, resulting in prohibitive economic and temporal costs. We therefore adopt fixed time steps ($t = 0.333$ s in this study) to replace grid-by-grid movement. Within each time step, each agent completes displacement corresponding to its movement speed and is assigned to the cell it occupies. The selection of time step t involves balancing multiple factors: excessively large t values reduce the real-time responsiveness of environmental updates and agent interactions, while overly small t values increase computational overhead and LLM dialogue frequency. The specific value of $t = 0.333$ s used here was chosen as it approximates the time required for an agent at typical walking speed (~ 1.5 m/s) to cross one grid cell (approx. 0.5 m), thereby linking the decision update rate to the spatial resolution of the environment. By updating agent states in cellular automata at fixed time intervals, we effectively reduce LLM dialogue requirements while simulating agents' capacity to formulate new movement strategies based on environmental changes. This approach preserves the discrete nature of cellular automata while maintaining the continuity of individual mobility. The pseudocode for this process is shown in Fig. 6.

Furthermore, in traditional cellular automata, objects with spatial semantics often correspond to a single cell, causing multiple evacuating individuals to move towards the same cell. This not only exacerbates cell competition but also makes the individuals' movement paths unrealistic. Therefore, in our study, an object containing spatial semantics is expressed through multiple cells, any of which can serve as a destination for the Agent's movement. To address cell preemption conflicts caused by simultaneous agent arrivals at target cells (with identical timestamps within precision), the proposed model employs a randomized allocation mechanism for conflict resolution: Random numbers are generated to determine priority, where the selected agent gains exclusive access to the target cell while the other agent is dynamically redirected to the nearest available adjacent cell.

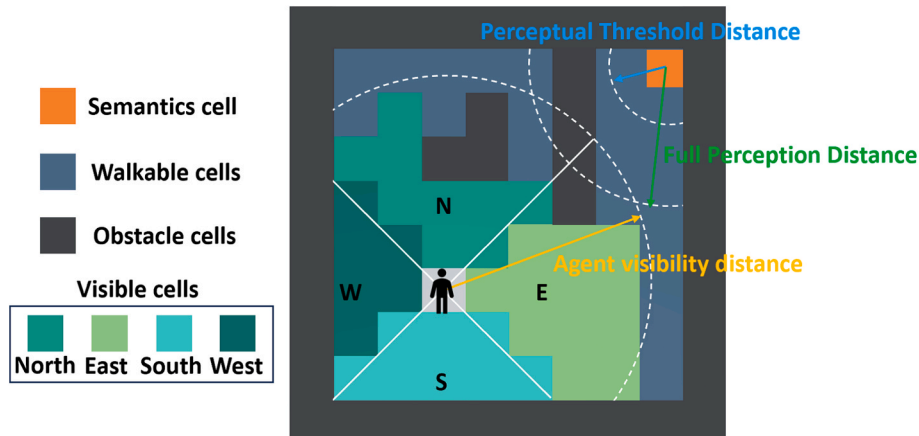


Fig. 4. Environment perception based on cellular automata.

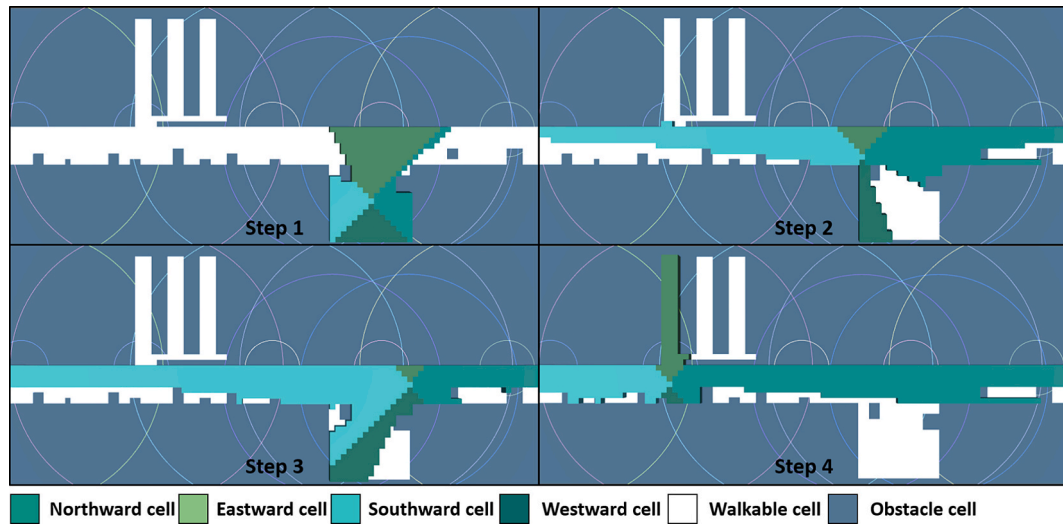


Fig. 5. Calculation of cell occlusion relationship.

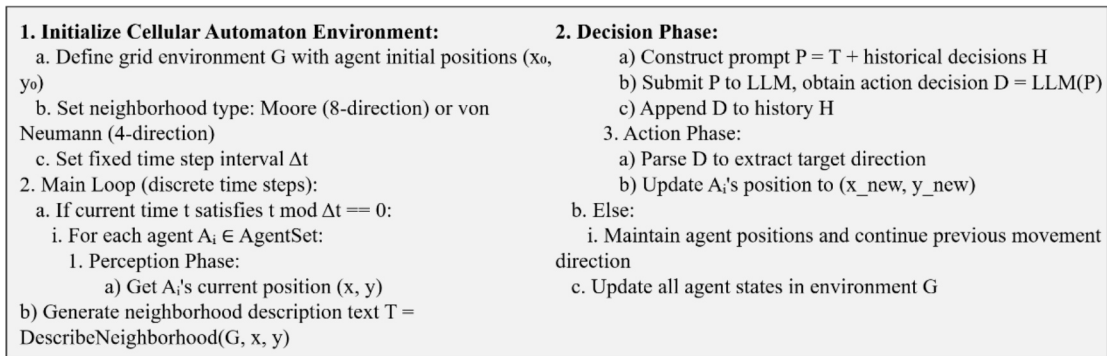


Fig. 6. Pseudocode for agent's action.

3.3. Multi-agent interaction behavior

In multi-agent systems for fire evacuation simulations, dynamic interactions among artificial intelligence agents (Agents) constitute a critical component for simulating realistic behaviors. These interactions primarily involve spatial positioning, the dynamics of competition and cooperation, and social network structures. These elements collectively influence the decision-making processes of agents and significantly impact the overall effectiveness of evacuation simulations.

To simulate such interactions, we designed a communication module for each agent. This module enables agents to record and exchange critical information at each simulation step, primarily consisting of environmental perception data, which is subsequently used for effective communication with neighboring agents. During each simulation step, an agent may choose to communicate with nearby agents (within 3.6 m). The communication content comprises summarized descriptions of fire scenarios retrieved from the agent's short-term memory, while excluding long-term memories, decision-making processes, historical movement paths, and other sensitive data. Received response information is integrated into the agent's short-term memory for future decision-making.

To implement this process, we introduced a decision-making mechanism in the agent's logic. At each simulation step, agents can choose to move toward their target direction, initiate communication with surrounding agents, or accept incoming communication requests. Agents receiving communication requests may opt to respond, though doing so requires maintaining a stationary position during that simulation step. All inter-agent communications follow a standardized JSON format

containing agent ID, timestamp, and fire scenario descriptions prepared for transmission. In this study, this module primarily facilitates fire-related information exchange between agents, simulating behavioral responses under conditions where agents lack direct visual perception of flames and smoke. While these descriptions provide a basis for more detailed content analysis in the future, the focus of this study is on the occurrence of communication and its impact on the basic transmission of fire information.

4. Experiment and result

4.1. Experiment design

4.1.1. Construction of a three-dimensional experimental scenario

This study utilized a LiDAR scanner (LiGrip H120) to scan a real scene and constructed a virtual fire evacuation scenario based on point cloud data and three-dimensional reconstruction results. We selected the first floor of a shopping mall as the experimental area, covering approximately 16,545 square meters. The point cloud data obtained from LiDAR scanning were processed using ContextCapture v10.20 for three-dimensional reconstruction, and a cellular automaton of 502*206 cells was constructed based on the reconstruction results. Given the total scanned area of approximately 16,545 square meters, each cell in this 502x206 grid represents a physical area of roughly 0.4 m x 0.4 m. This defined spatial scale is used consistently throughout the simulation for translating real-world parameters, such as agent perception distances and movement speeds, into the cellular automata framework. Finally, spatial semantics were manually added to the cells based on the

panoramic video and point cloud data from the 3D laser scanning, setting spatial semantic information for 62 objects in the evacuation environment (such as exits, evacuation signs, etc.), as shown in Fig. 7.

4.1.2. Simulation of flames and smoke

In this study, we used PyroSim 2022 software to simulate the spread of flames and smoke under controlled conditions. PyroSim is a graphical interface for the Fire Dynamics Simulator (FDS), capable of detailed simulation of flames, smoke, and heat transfer phenomena. The burning object was set to fabric, with a heat release rate of 600 kW/m² per unit area, and the burning area was set to 5 square meters. Two exits were kept open as ventilation ports, with an area of 3 m x 4 m and an air flow rate of 0.5 m/s. Under these conditions, the simulation results of flames and smoke spread from PyroSim were imported into Unity 2021.3.11 for visualization and cell description, as shown in Fig. 8.

We categorized visibility into seven levels based on ISO 13571 (ISO, 2012), establishing corresponding prompt messages to describe the Agent's surrounding environment. This refined gradation system enables LLMs to generate differentiated decisions through variations in descriptive vocabulary, as detailed in Table 1. Objects beyond visibility thresholds were not included in environmental prompts, though explicit notifications were provided in system messages indicating their obstruction by smoke. This study enables agents to perceive and respond to complex environments by converting environmental parameters such as smoke into structured text inputs for the LLM. The method is also applicable to other environmental factors, providing a scalable interface for building a general decision-making framework.

4.1.3. Evacuees and initial paths

In the evacuation scenario, 10 Agents were set, among which 5 Agents (numbered 1, 3, 5, 6, 9) moved along an initial path at the start of the experiment until they autonomously generated their first evacuation decision. The other Agents remained stationary, waiting to generate their first autonomous decision, as shown in Fig. 9. Considering Agents move towards semantic cells as their destination, the movement process is facilitated by Unity's Navmesh component, with walking speeds set between 1 m/s and 1.7 m/s, and running speeds between 2.2 m/s and 4.2 m/s. Agents begin to evacuate under any of the following conditions: 1. Flames and smoke are within the Agent's perception range; 2. After communicating with other Agents and obtaining information about the fire; 3. Upon hearing the alarm, initiate evacuation procedures; the alarm will sound 120 s after a fire breaks out.

Moreover, basic attribute information and context backgrounds were added to each Agent. This information was submitted in the form of

System-Prompts, as shown in Table 2.

4.1.4. Experimental and control group setup

Currently, many organizations and companies have developed both open-source and proprietary LLMs, which exhibit differences in text comprehension, generation abilities, and generalization capacities. Among the factors influencing the performance of LLMs, the number of model parameters is significant. Therefore, to verify whether our research method is universally applicable across different LLMs, we selected LLMs of various scales for the experimental group, as shown in Table 3.

Although some organizations have not disclosed the parameter counts of their LLMs, based on evaluations and estimates by other researchers, these models are believed to possess larger parameter counts and have thus been adopted in our study.

As a comparison, we developed an evacuation program in Unity as the control group, where Agents move only towards the nearest exit, and all Agents' Response Time is set to 120 s. The movement of Agents in the control group employs the A* algorithm, hence naming the control group as Control Group-A* (CG-A*).

4.2. Results

4.2.1. Evacuation duration and paths

In the outcome experiment, we randomly selected a group from each experimental group to present the final evacuation paths of the Agents, as shown in Fig. 10.

We calculated the path length based on the number of cells each Agent passed through in the experiment and summed up the movement distances for all Agents in each group. After conducting the Shapiro-Wilk test and Levene's test on the data from each experimental group, the results indicated that the data from the four experimental groups generally followed a normal distribution but did not satisfy the homogeneity of variance test. Therefore, we used the T-test to compare the differences between groups, the significance level is set at 0.05, as shown in Table 4.

Considering the significant differences in evacuation paths among Agents due to their diverse background settings in each group, comparing the same Agents across different groups yields more meaningful insights. The results of the T-test comparing Agents across different groups are shown in Fig. 11.

Moreover, the movement speeds of Agents were variable, affecting the overall evacuation duration for each group. We compiled the remaining number of evacuees during the evacuation process for each group, as illustrated in Fig. 12.

The CG-A* group moved along the shortest path, thus taking the least actual time. T-tests among the experimental groups revealed no significant difference between EG-1 and EG-2 ($T = -0.306$, $p = 0.763$) or between EG-3 and EG-4 ($T = -0.247$, $P = 0.808$). However, significant differences were found between EG-1 and EG-2 when compared with EG-3 and EG-4 ($P < 0.05$), mirroring the variations in evacuation duration with differences in evacuation path lengths.

4.2.2. Evacuation decisions

During the evacuation process, various factors ultimately influence the evacuation distance and time, all of which are expressed in LLM's responses in each round. Therefore, we analyzed LLM's decisions in each round to explore the factors specifically affecting the evacuation outcomes. We collected statistics on the Agents' changes in evacuation direction, the number of interactions with other Agents, whether they sought companions based on background information, and whether they passed through dangerous areas (cells covered by smoke).

In the evacuation process, we adopted an Agent-centered spatial orientation, describing the locations of different objects in terms of cardinal directions. Changes in evacuation direction often imply more evacuation time. Considering the different initial positions and

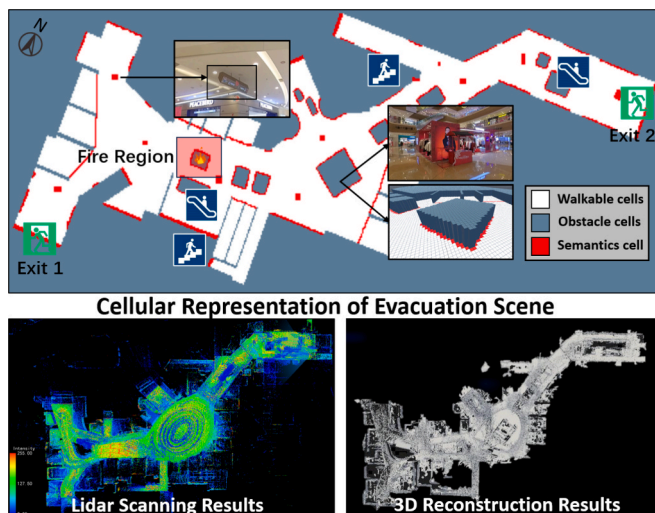


Fig. 7. Construction of evacuation scenario and cell setup.

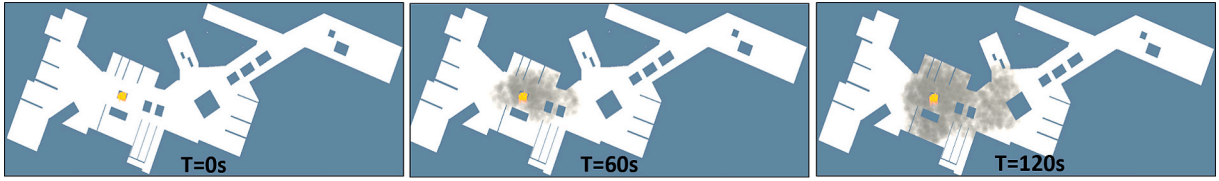


Fig. 8. Spread of smoke.

Table 1
Visibility grading and prompt settings.

No.	Visibility Range (m)	Level	Prompt
1	=0	Extremely Low Visibility	The smoke is so dense that visibility is completely obstructed, making it impossible to discern any objects or directions.
2	(0,0.5]	Very Low Visibility	The smoke is very heavy, only allowing the sight of objects that are very close, with no discernible details of anything further away.
3	(0.5,1.5]	Low Visibility	The density of the smoke is high, significantly limiting visibility, but still permitting the identification of some nearby objects and outlines.
4	(1.5,2.5]	Moderate Visibility	The density of the smoke is moderate, with distant objects starting to blur, but nearby objects can still be seen relatively clearly.
5	(2.5,5]	Relatively High Visibility	The smoke is relatively light, enabling the sight of more distant objects, although the vision may still be slightly obstructed.
6	(5,10]	High Visibility	While smoke is present, its impact on visibility is minimal, hardly hindering the normal field of view.
7	>10	Extremely High Visibility	The smoke is almost invisible or does not affect visibility at all, allowing for a clear view of objects at all distances.

background information of different Agents, we also compared the number of decision changes among the same Agents across different groups, as shown in Fig. 13.

The CG-A* group's Agents changed their direction of movement the least during evacuation ($M = 1.6 \pm 1.20$). In the experimental groups, there was no significant difference between EG-1 ($M = 4.27 \pm 2.68$) and EG-2 ($M = 4.43 \pm 2.74$) ($T = -0.114$, $p = 0.912$), nor between EG-3 and EG-4 ($T = -0.492$, $p = 0.628$). For Agent 1, the number of direction changes in EG-1 and EG-2 was higher than in EG-3 and EG-4, due to Agent 1's initial information including seeking friends in the mall. This process resulted in more direction changes in EG-1 and EG-2, while similar decisions in EG-3 and EG-4 were fewer in number.

Furthermore, if Agent 1 decided to seek companions, it needed to

pass through dangerous areas. We counted the number of times Agent 1 in different groups decided to seek its companion (Agent 8) under the known fire occurrence and the number of times all Agents passed through dangerous areas, as shown in Fig. 14.

In EG-1 and EG-2, Agent 1 decided to seek Agent 8 a total of 5 times out of 10 tests in EG-1 and EG-2, 3 times in EG-3, and 2 times in EG-4. Of these, EG-1 had 4 successful searches for Agent 8, EG-2 had 5, EG-3 had 1, and EG-4 had none. During the search, Agent 1 passed through dangerous areas, and other instances of passing through dangerous areas were caused by Agent 4 after starting the evacuation.

Finally, we compiled the communication behaviors among Agents in each group. Considering that not all Agents engaged in communication with other Agents during the evacuation process, we counted the communication occurrences for each Agent individually, as shown in

Table 2
Setting of agent background information.

No.	Name	Gender	Age	Role	Social relations
1	Alex	Male	35	Customer	Preparing to go to the clothing store and meet his friend Mike.
2	Sarah	Female	44	Food store employee	Working alone in the store.
3	Emily	Female	24	Customer	Preparing to go to the toy store to purchase toys.
...					

Table 3
Information on LLMs in the Experimental Group.

Group (EG = Experiment Group)	Model Name	Organization	Parameters (Billion)	Open-Source
EG-1	ChatGPT 4.0	OpenAI	>1750	No
EG-2	ERNIE-Bot 4.0	Baidu	>100	No
EG-3	Llama-2-70B-Chat	Meta	70	Yes
EG-4	ChatGLM2-6B-32 K	THUDM	6	Yes

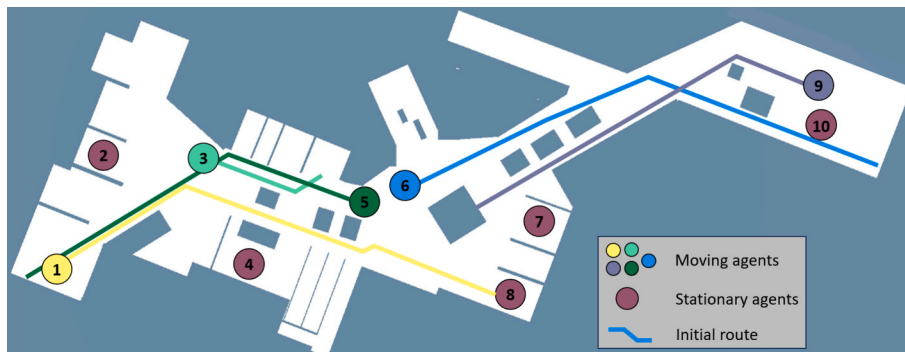


Fig. 9. Initial positions and paths of agents.

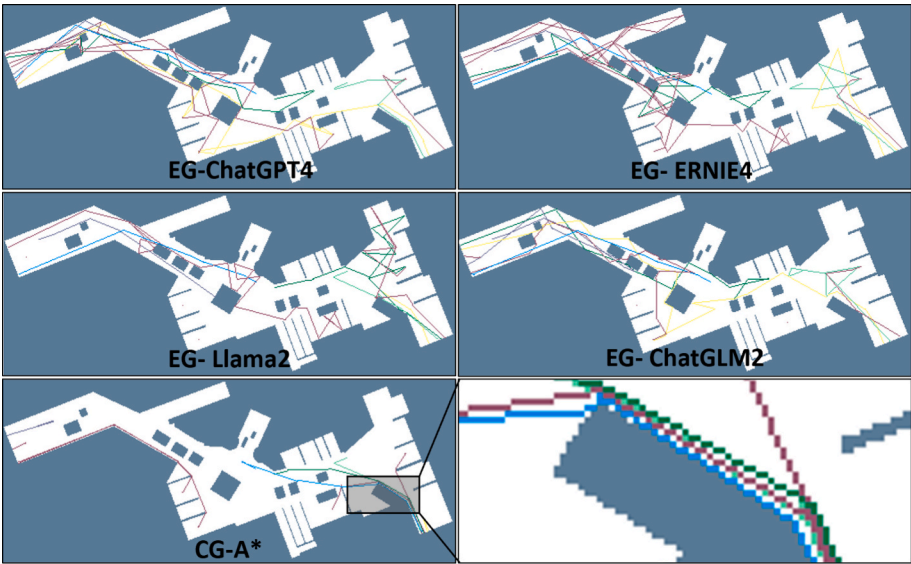


Fig. 10. Evacuation paths in part of the experiment.

Table 4
Evacuation distance and time for each group.

Group	Evacuation Distance (cell)	Evacuation Time (s)
EG-ChatGPT4	2829.3 ± 224.58 ^a	71.03 ± 9.56 ^a
EG-ERNIE4	3080.4 ± 182.27 ^b	75.83 ± 7.91 ^a
EG-Llama2	3811.1 ± 168.61 ^c	109.16 ± 7.17 ^b
EG-ChatGLM2	3745.6 ± 114.46 ^c	115.01 ± 6.38 ^c
CG-A*	1181.0 ± 0.0 ^d	142.28.0 ± 0.0 ^d

Fig. 15.

The results showed no difference between EG-1 ($M = 8.4 \pm 1.837$) and EG-2 ($M = 8.5 \pm 2.068$) ($T = -0.114$, $p = 0.910$), nor between EG-3 ($M = 6.2 \pm 1.751$) and EG-4 ($M = 4.9 \pm 1.523$) ($T = 1.771$, $p = 0.093$). In EG-1 and EG-2, Agents 1, 3, and 5 frequently decided to communicate with other Agents, accounting for 58.33 % (EG-1) and 62.3 % (EG-2) of the total, whereas in other groups, they accounted for 23.43 % (EG-3) and 14.28 % (EG-4). Conversely, in EG-3 and EG-4, Agents 4, 7, and 8 contributed the most communications, accounting for 65.62 % (EG-3)

and 73.46 % (EG-4), while these three Agents contributed only 27.38 % (EG-1) and 16.47 % (EG-2) in other groups. There are differences in communication frequency among the experimental groups and agents, mainly used to transmit critical environmental information such as fire notifications. For example, in group EG-3, agent 2 significantly reduced evacuation time after receiving a fire notification.

5. Discussion

5.1. Evacuation efficiency

For Agents, emphasis is placed on their exploration of and decision-making in the environment, whereas traditional Agent evacuation models stress the homogeneity and orderliness of individual behaviors. Traditional Agent evacuation models, such as the Social Force Model (Helbing et al., 2000) and Cellular Automata Model (Pelechano & Badler, 2006), typically assume that all individuals follow the same behavioral rules, such as moving towards exits, avoiding obstacles, and maintaining distance from others. Although this assumption of

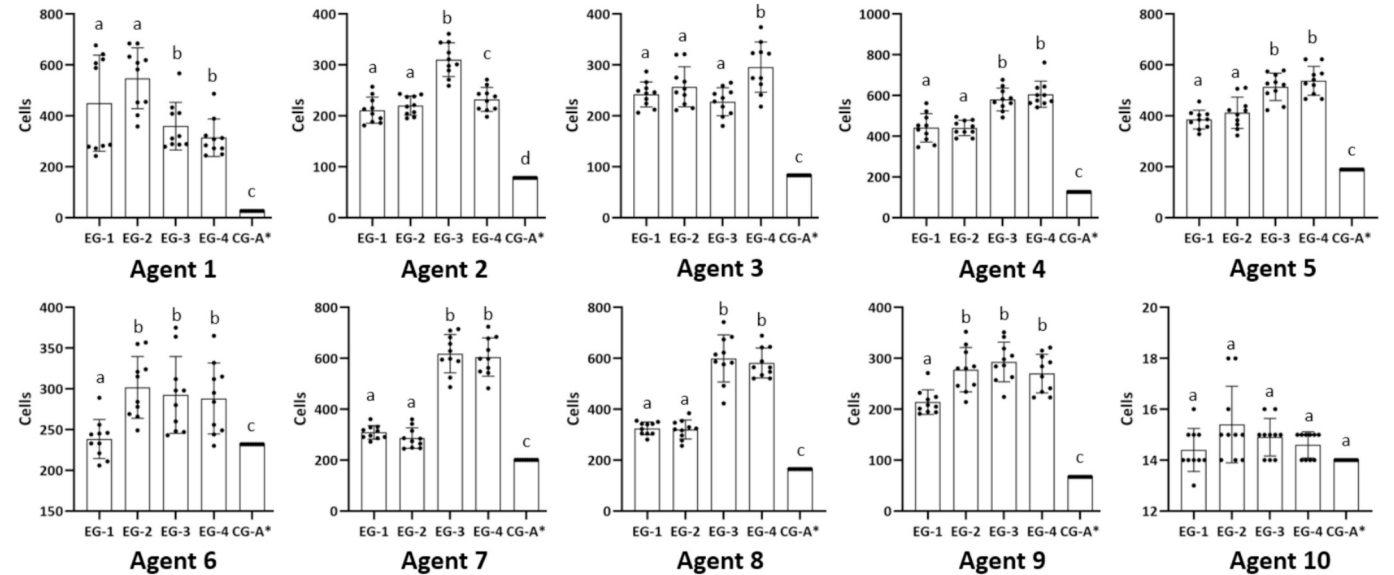


Fig. 11. Comparison of agent evacuation distances.

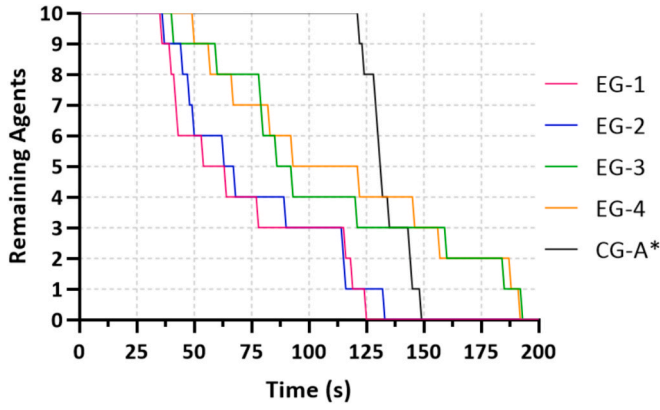


Fig. 12. Remaining number of evacuees.

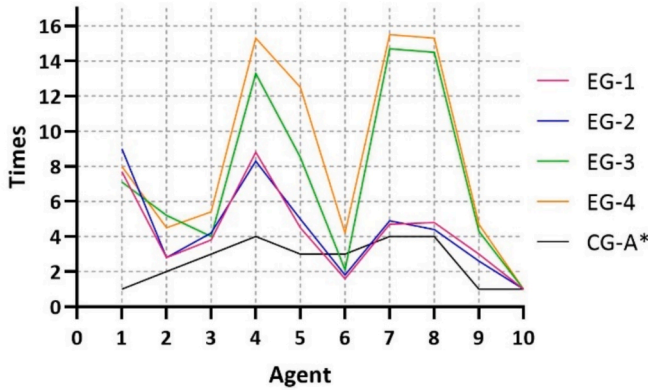


Fig. 13. Number of changes in agents' evacuation decisions.

homogeneity helps improve the computational efficiency and controllability of the model, it also limits the diversity and adaptability of individual behaviors. In real evacuation processes, individuals often make different decisions based on their own attributes, experiences, and environmental conditions, and it is this diversity that is key to enhancing the realism and effectiveness of evacuation.

Compared to agent models utilizing deep reinforcement learning (Hou et al., 2022) or genetic algorithms (Zhai & Feng, 2022), LLM-based evacuation agent models demonstrate superior capabilities in capturing the heterogeneity and dynamic nature of individual decision-making, while enabling more direct integration of human knowledge for reasoning and decision-making. Through environmental perception and comprehension, LLM-driven agents dynamically adjust evacuation

strategies in response to real-time conditions, exhibiting enhanced exploratory and adaptive behaviors. This exploratory behavior leads to extended navigation time in experimental groups, ultimately reducing evacuation efficiency. However, such exploration is commonly observed in fire evacuations, particularly among evacuees unfamiliar with the environment, who frequently modify their exploration patterns and wayfinding strategies (Cao et al., 2019). The diverse evacuation paths exhibited by different LLM-driven agents in this study (as shown in Fig. 10) also intuitively corroborate this, aligning with the characteristics of spontaneous wayfinding behaviors described in literature when individuals face uncertainty.

Notably, larger-scale models demonstrate superior wayfinding efficiency, with EG-1 and EG-2 significantly outperforming EG-3 and EG-4 across multiple metrics, while similarly scaled LLMs exhibit comparable performance. Except for Agent 6 showing significant path variation between EG-1 and EG-2, other agents maintained similar evacuation path lengths. This discrepancy arises because Agent 6 followed its initial route until blocked by building structures and triggered fire alarms, failing to fully acquire fire outbreak information from other agents – a situation entirely driven by procedural randomness. Crucially, this randomness differs fundamentally from real-world evacuation behavior variability: it stems from textual output stochasticity in small-scale LLMs, contrasting with decision-making randomness observed in large-scale LLM evacuation simulations.

5.2. Evacuation behaviors

Differences in evacuation behaviors are a significant factor leading to variations in evacuation outcomes. Except for the control group, which only considered the shortest path, other groups exhibited complex and variable behaviors during evacuation. Firstly, whether Agents were informed of the fire outbreak by other Agents significantly impacted the overall evacuation time. For example, in EG-3, Agent 2's evacuation time without notification from other Agents was 158.19 ± 10.90 s, while with notification, it dropped to 82.33 ± 36.96 s. Experimental results indicate that notifying other Agents about the fire is a random behavior; even under the same LLM parameters, Agent behaviors showed variability, but larger-scale models reduced this randomness. The coefficient of variation for communication among Agents in the experimental groups was calculated as EG-1: 0.208, EG-2: 0.231, EG-3: 0.268, EG-4: 0.295.

Moreover, the background setting of Agents significantly affected their evacuation behaviors. In the prompts, we did not strictly require seeking Agent 7 for evacuation, but the decision to find Agent 7 was made by Agent 1 in all experimental groups, resembling actual evacuation scenarios (Dang et al., 2024). However, cases of giving up the search occurred in EG-3 and EG-4, possibly due to three reasons: firstly, increasing context in prompts could lead LLM to miss critical

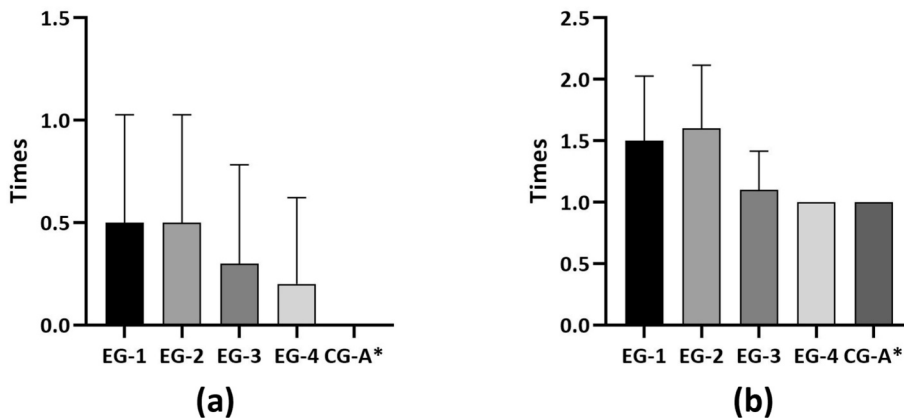


Fig. 14. Times of searching for other agents (a) and Times of passing through smoke cell (b).

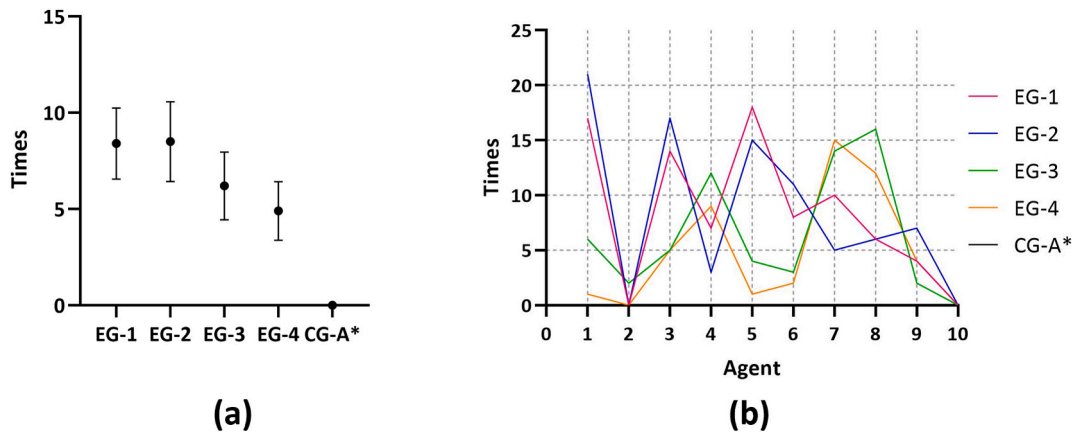


Fig. 15. Times of group communication (a) and Times of individual agent communication (b).

information due to difficulty in capturing focus. Secondly, abandoning previous decisions could lead Agents to give up searching for companions midway. Lastly, reaching Agent 7's location required Agent 1 to pass through dangerous areas, and in some tests, Agent 1 chose to find the nearest exit after weighing the options. Changes in decisions not only occurred while seeking companions but also in searching for exits, often when Agents continued exploration in a direction without finding the desired target. This behavior aligns closely with the classic 'affiliative behavior' phenomenon in evacuation behavior studies, where individuals in crisis situations tend to seek out and evacuate with relatives or friends rather than solely pursuing the quickest personal escape route, as discussed in studies such as Fischer et al. (1995).

Frequent changes in decisions led to increased evacuation times. We calculated the Pearson correlation coefficient between the number of decision changes and evacuation distance: EG-1: 0.820, EG-2: 0.604; EG-3: 0.485, EG-4: 0.571. However, changes in decisions did not necessarily imply more efficient exploration, as some tests found Agents wandering between multiple objectives. We assessed the effectiveness of decision changes in each group using the ratio of evacuation distance to the number of decision changes, finding EG-1: 66.61 ± 4.54 , EG-2: 70.60 ± 6.35 , EG-3: 51.07 ± 2.47 , EG-4: 43.49 ± 2.29 . EG-2 performed best, with each decision change covering approximately 70 cells, whereas EG-4 had the most changes but the least effectiveness. Notably, in real evacuation scenarios, changes in decisions relate to environmental changes (Zheng et al., 2017), but in this study, evacuation decision changes were linked to LLM's memory and reasoning capabilities.

5.3. Reasoning-based evacuation decision

The core characteristic of LLM-driven agent evacuation models lies in their autonomous reasoning capabilities based on real-time information, rather than reliance on predefined rules. At the individual level, agents exhibit behavioral diversity in evacuation scenarios – even with identical parameters, different agents display behavioral variations. However, at the group level, behavioral patterns stabilize under the constraints of a unified perception-action framework, explaining the approximate distribution patterns of communication frequencies across experimental groups.

Experimental findings reveal that individual decision variations under identical parameters originate from LLMs' implicit reasoning capabilities, though these capabilities demonstrate significant limitations: In explicit reasoning scenarios (e.g., "exit located west" prompts), EG-3/EG-4 group agents still exhibit non-westerly movement choices (with occurrence frequencies significantly higher than EG-1/EG-2 groups). This suggests smaller-scale LLMs may fail to accurately parse critical information (Wei et al., 2022), leading to more invalid/erroneous decisions. When confronted with ambiguous or incomplete

environmental information, even large-scale LLMs may generate uncertain judgments. Implicit reasoning capability directly influences model generalization performance, serving as a necessary but insufficient condition for LLMs to simulate realistic human behaviors. All experimental LLMs autonomously developed fire notification behaviors (not explicitly required in prompts), with large-scale models (EG-1/EG-2) demonstrating more proactive responses, particularly in actively approaching stationary agents for warnings. This behavior stems from implicit reasoning that "stationary status may indicate undetected fire hazards." However, such reasoning might be influenced by ethical biases in training data, potentially causing over-reliance on idealized social norms.

Therefore, while larger-parameter models demonstrate superior performance in critical decision-making and behavioral simulation, their limitations in processing dynamic information and overcoming training biases remain significant concerns. These factors could substantially impact model effectiveness in practical evacuation scenarios (Fischer et al., 1995; Efferson et al., 2024).

The human-like characteristics demonstrated by LLMs in implicit reasoning tasks stem from their ability to capture and learn from human knowledge and behavioral patterns in the training data. Nevertheless, due to the potential introduction of specific ethical and moral biases by different developers during the training process, the specific performance of each LLM may vary. Overall, the formation of LLMs' human-like reasoning capabilities relies on the coverage of human knowledge in the training data and the guidance of developers' value orientations, which are also important premises for their application in agent behavior simulation.

6. Conclusion and future work

This study introduced a multi-Agent fire evacuation model based on LLM, endowing Agents with personalized memory, cognition, and decision-making capabilities through LLM. By characterizing the evacuation environment through a combination of spatial semantics and cellular automata, Agents can perceive, decide, and move within a unified spatiotemporal framework. Experimental results demonstrated that this framework enables LLM-Agents to complete fire evacuation tasks, exhibiting exploratory and adaptive behaviors consistent with real scenarios. Additionally, LLMs of different scales exhibited variations in the diversity and randomness of Agent behaviors, with larger-scale LLMs generating more consistent and efficient evacuation strategies.

This study has several limitations that indicate clear directions for future research. First, although the model incorporates heterogeneous agent attributes, it emphasizes collective evacuation behaviors under multi-factor coupling, with less focus on the independent impact of individual variables (e.g., age, social role). Future work should prioritize

integrating more diverse agent populations—considering differences in age, mobility, social responsibilities (e.g., staff, caregivers), and environmental familiarity—to improve the ecological validity and capture nuanced evacuation dynamics in real-world public settings. Moreover, the lack of quantitative validation against real-world data limits behavioral fidelity to qualitative assessments. Second, the small agent population ($N = 10$) and limited simulation repetitions—constrained by the high computational cost of LLM-driven approaches—limit statistical robustness and generalizability. Third, the model uses simplified evacuation triggers and does not explicitly differentiate cognitive and behavioral phases within pre-movement time (e.g., detection, recognition), which may affect the accuracy of delay modeling. Lastly, the decision-making logic of LLM-based agents lacks interpretability; developing tools for traceability and visualization is essential to enhance transparency and trust. Addressing these issues will be key to advancing LLM-based evacuation modeling toward reliable real-world applications in urban resilience.

Data and codes availability statement: The data and codes that support the findings of this study are available in [figshare.com](https://figshare.com/s/333e166317fa3a6afd90) with the identifier <https://figshare.com/s/333e166317fa3a6afd90>.

CRediT authorship contribution statement

Pei Dang: Methodology, Software, Writing – original draft. **Jun Zhu:** Data curation, Supervision. **Weilian Li:** Validation, Data curation. **Yakun Xie:** Resources, Software. **Heng Zhang:** Funding acquisition, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. Jun Zhu reports financial support was provided by Southwest Jiaotong University.

Acknowledgements

This research was funded by the National Natural Science Foundation of China [Grant Nos. 42271424, 42171397, 42201445], the Open Project Fund of National Key Laboratory of Intelligent Parallel Technology [SHJJ2024013], the Open Project Fund of National Engineering Research Center of Digital Construction and Evaluation Technology of Urban Rail Transit [No. 2024sys015], and the Natural Science Foundation of Tianjin [No. 23JCQNJC01130].

Data availability

Data will be made available on request.

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